



AUTUMN END SEMESTER EXAMINATION-2025

3rd Semester B.Tech

DATA STRUCTURES / DATA STRUCTURE AND ALGORITHMS

CS21001 / CS 2001

(For 2024 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes

Full Marks: 50

Answer any FIVE questions.

Question paper consists of two SECTIONS i.e. A and B.

Section A is compulsory.

Attempt any Four question from Sections B.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

- (a) Determine the time complexity of the following code snippet and specify the final value of count.

```
int n=14, count=0;
for(int i=n; i>0; i/=2) {
    for( int j=0; j<i; j++)
        count++;
}
```

- (b) An array `arr[2...6, -2...6]` stores each element using 4 bytes of memory. If the base address is 2001, determine the address of `arr[5][3]` in column-major order.
- (c) Explain why stack overflow can occur in a static implementation of a stack but not in a dynamic implementation.

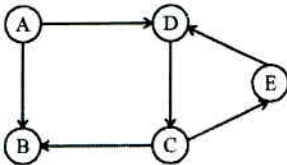
- (d) Represent the following sparse matrix using a space-efficient storage scheme i.e. triplet representation:

[0 0 5 0]

[0 8 0 0]

[0 0 0 6]

- (e) Present the Adjacency Matrix Representation of the following graph.



- (f) Write a C function to count the number of non-leaf nodes in a binary tree.
- (g) Construct an expression tree for the expression $((a + b * c) + (d * e + 1) * g)$.
- (h) Given the following inorder and preorder traversals of a binary tree, construct the binary tree and determine its postorder traversal.

Inorder: D B E A F C

Preorder: A B D E C F

- (i) What is the content of the array after the second pass of Bubble Sort on [5, 3, 8, 4, 2]?
- (j) Consider a hash table with 7 slots and a hash function $h(k) = k \bmod 7$. The following keys are inserted in order: 15, 38, 8, 5, 20, 33, 14, 28. Draw the final contents of the hash table when collisions are resolved using chaining.

SECTION-B

2. (a) Write a C function to add two polynomials. Use singly linked lists to represent the polynomials, and illustrate both their linked representations and the result after addition.

[5]

$$P1: 5x^3 + 4x^2 + 7x + 3$$

$$P2: 6x^2 + 5$$

- (b) Write a C-like function that reverses the values present at the odd positions of an array containing n elements. [5]
3. (a) Write a pseudocode or C-like function to determine the intersection point (the data value where they converge) of two singly linked lists that merge at a common node. [5]
- (b) Write a pseudocode or C-like function to split a given circular doubly linked list into two almost equal circular doubly linked lists, without counting the total number of nodes in the list. Use appropriate pointer manipulation to determine the splitting point. [5]
4. (a) State the algorithm (with clear pseudocode) for evaluating a postfix expression using an appropriate data structure. Using the algorithm to evaluate the following postfix expression step-by-step, showing the contents of the stack after processing each token: $A B + C * D E F G * + / + H -$. Use the values: $A = 4, B = 8, C = 2, D = 5, E = 6, F = 9, G = 1, H = 3$. [5]
- (b) A Circular Queue of size 5 (with C-style indexing: 0 to 4) currently holds three elements: 10, 20, and 40, where Front (F) = 2 and Rear (R) = 4. Determine the updated values of Front (F) and Rear (R) after inserting 50 and 60. Explain what occurs if an attempt is made to insert 30 at this stage. Next, delete two elements from the queue and then insert 70, 80, and 90. Present each operation with a step-by-step diagram of the queue, clearly showing the state of the queue and the values of F and R after every operation. [5]
5. (a) Explain the concept of the balance factor in an AVL tree. Then, insert the following elements: 2, 5, 4, 6, 7, 9, 8, 3, 1, 9, into an initially empty AVL tree, ensuring [5]

that the tree is balanced after each insertion. Provide a step-by-step explanation of the insertion and balancing process.

- (b) Write a pseudocode or C-like function to perform Depth First Search (DFS) traversal on a graph. Demonstrate the step-by-step execution of the DFS traversal starting from vertex A, considering the lexicographic order of adjacent vertices, for the given graph G: [5]

Vertex	Adjacent Vertices
A	B, D
B	A, C, E
C	B, F
D	A, E
E	B, D, F
F	C, E

6. (a) Write a pseudocode or C-like function to perform Insertion Sort on an array and display both the sorted array and the total number of element shifts (data movements). Also, demonstrate the step-by-step execution of the algorithm for the given array: [9, 5, 1, 4, 3]. [5]
- (b) Consider a hash table of size 11 and a hash function defined as: $h(k) = (3k+5) \bmod 11$. Using this hash function, insert the following sequence of keys into the hash table: [12, 44, 13, 88, 23, 94, 11, 39, 20, 16, 5]. Assume that linear probing is used for collision resolutions. Construct the final hash table after all insertions. Determine the total number of collisions that occur during the insertion process. [5]
